

Question number	Answer	Additional guidance	Mark
4(a)	7  and 8  and 9 		(1)

Question number	Answer	Additional guidance	Mark
4(b)(i)	A description that includes the following points: - gene is a sequence of bases coding for a {(poly)peptide / sequence of amino acids} + gene for bitter-taste receptor (1) - allele is a {version of the gene / information coded by the gene} + {tasting / not tasting} (receptor) (1)	ACCEPT protein NB Two correct definitions but no examples = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	A description that includes the following points: - genotype is the combination of alleles + {TT / tt / Tt} (1) - phenotype is the {expressed / observable} {characteristic / trait / feature} + {taster / non-taster}	ACCEPT any pair of letters ACCEPT characteristic that can be measured {tasting / not tasting} (receptor) NB Two correct definitions but no examples = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
4(c)	An explanation that includes the following points: - dominant allele is (having a receptor that) can taste bitter taste (1) - because individuals 1 and 2 were tasters who had children of both phenotypes (1)	ACCEPT children who could taste and children who could not taste	(2)

Question number	Answer	Additional guidance	Mark
4(d)	An explanation that includes the following points: - because individual 6 was a female non-taster (1) - and her father was a taster (1)		(2)